Advanced Programming Practice · 21CSC203P

Unit 4 · Session 10 · SLO 2 — Solved Worksheet

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Unit â€“ 4 : Session â€“ 10 : SLO â€“ 2 | Client-Server Communication Using the socket Module

I. Steps to Create a Simple TCP Server

1. Create the socket: s = socket.socket(socket.AF_INET, socket.SOCK_STREAM) # IPv4 + TCP
2. Bind it to an address: s.bind(("localhost", 5000))
3. Start listening: s.listen(1) # backlog of 1 pending connection
4. Accept a client: conn, addr = s.accept() # blocks until a client connects
5. Exchange data: data = conn.recv(1024); conn.send(b"Hello client")
6. Close both sockets: conn.close(); s.close()

II. Minimal Working Example

• import socket
• s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
• s.bind(("localhost", 5000))
• s.listen(1)
• conn, addr = s.accept()
• print("Connected:", addr)
• conn.sendall(b"Welcome")
• conn.close(); s.close()